



Predicting NHL Expected Goals (xG):

A Machine Learning Approach
to Shot Quality

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The Challenge of Evaluating NHL Offense

- Hockey is a fluid, low-scoring sport
- Shot volume $\neq$ Shot quality
- xG quantifies the danger of every shot



Project Objectives



Defining the Analytics and Technical Goals

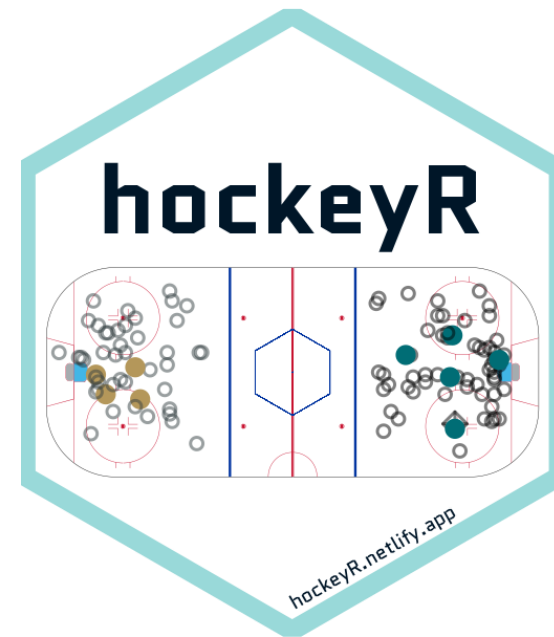
- Sports Analytics Goal:
 - *Objectively measure shot quality*
- Machine Learning Goal:
 - *Build a binary classification model*
 - Target Variable: *is_goal* (1 = goal, 0 = no goal)

Data Sourcing & Overview



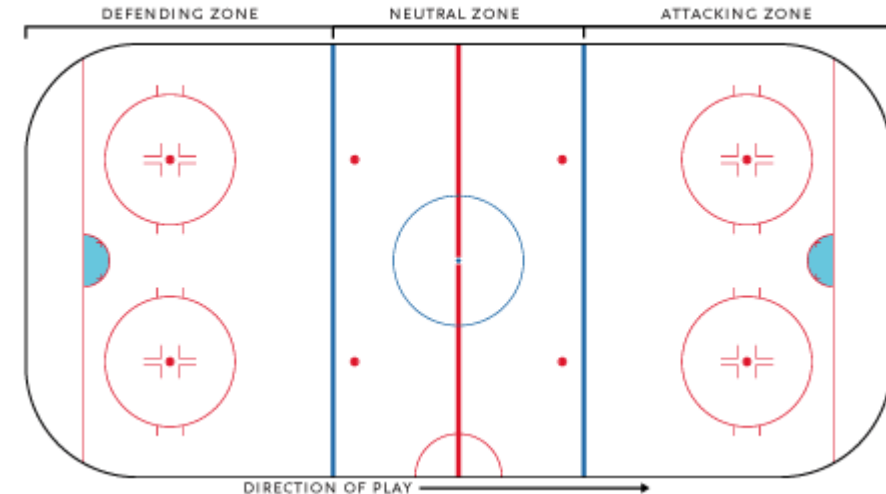
Five Seasons of NHL Play-by-Play Data

- hockeyR API
- 2018-2024 (excluding 20-21)
- Over 330,000 shots
- Features
 - *shot_distance*
 - *shot_angle*
 - *shot_type*



Standardizing the Hockey Environment

- Coordinate Folding
- Filtering
- Categorization
- Chronological Split

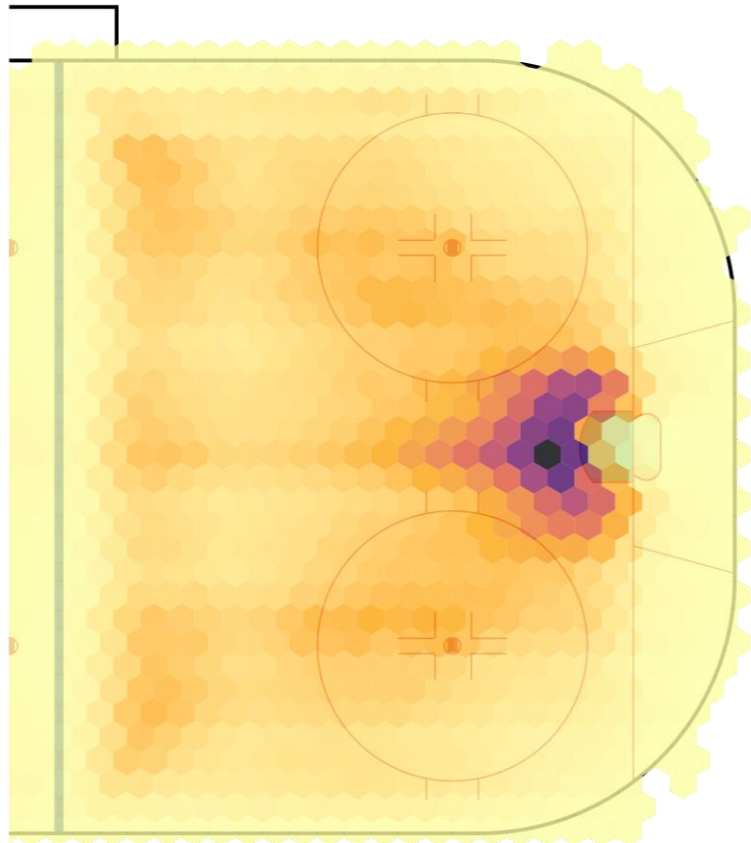


Exploratory Data Analysis



NHL shots are heavily concentrated in the high-danger slot

Unblocked shots from 2018-2024 (Excluding 2020-21)



Data: hockeyR | Map: sportyR

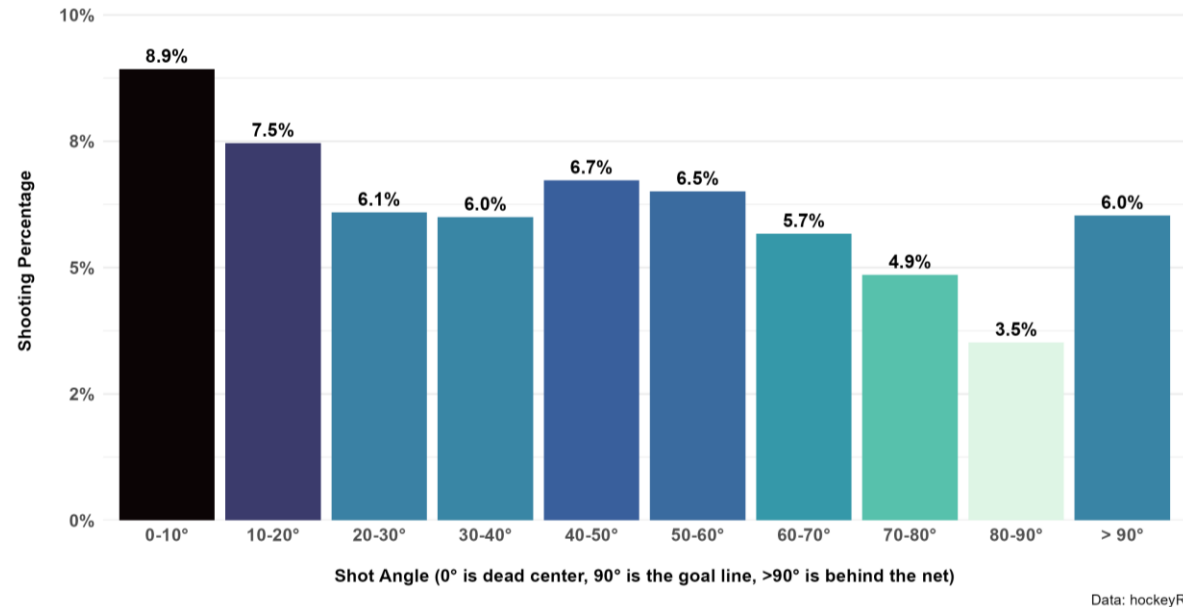
- Center + Low Slot
 - Angles
 - Point

Exploratory Data Analysis



Shooting percentage naturally fades as shots move away from dead center

Conversion rate by shot angle (2018-24, excluding 2020-21)

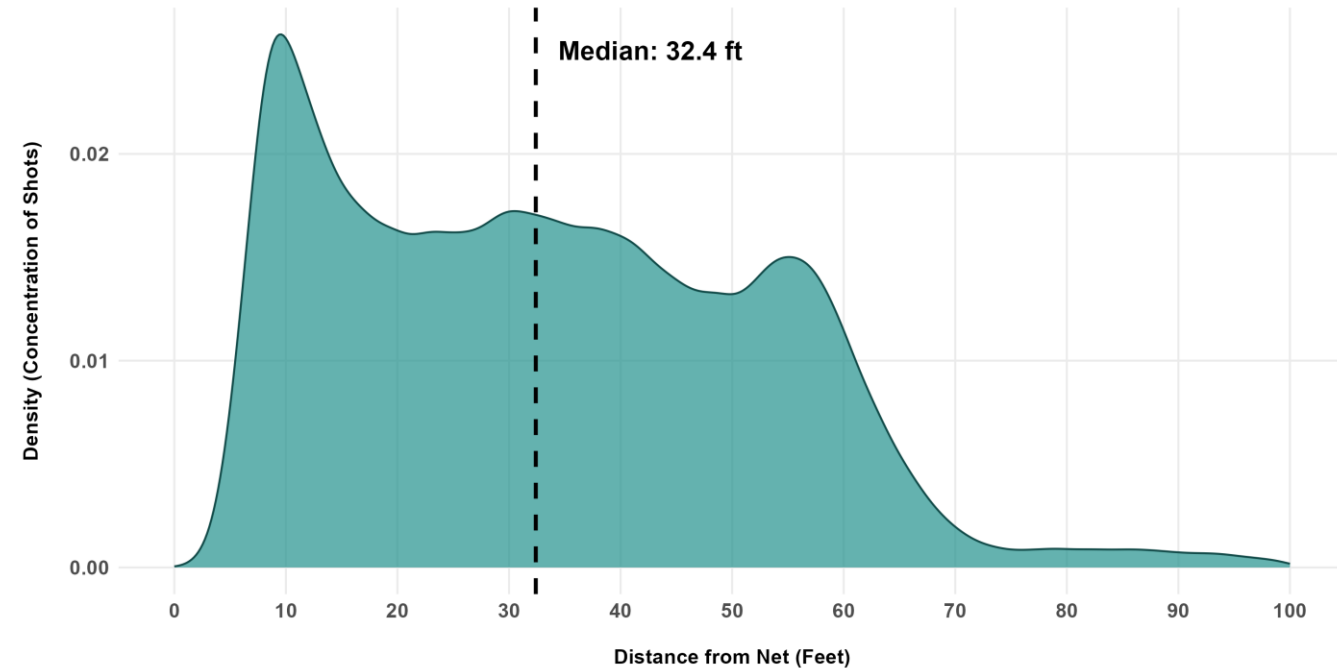


Shot Angle	Total Shots	Goals
0-10°	107635	9605
10-20°	86505	6459
20-30°	114230	6961
30-40°	104276	6252
40-50°	68580	4616
50-60°	43854	2854
60-70°	28408	1611
70-80°	18447	896
80-90°	8464	298
> 90°	7098	428

- 40-60 are cross ice one timers
- Sample sizes

Shot frequency drops dramatically beyond the faceoff circles (40 feet)

Density distribution of unblocked shot distances (2018-24, excluding 2020-21)



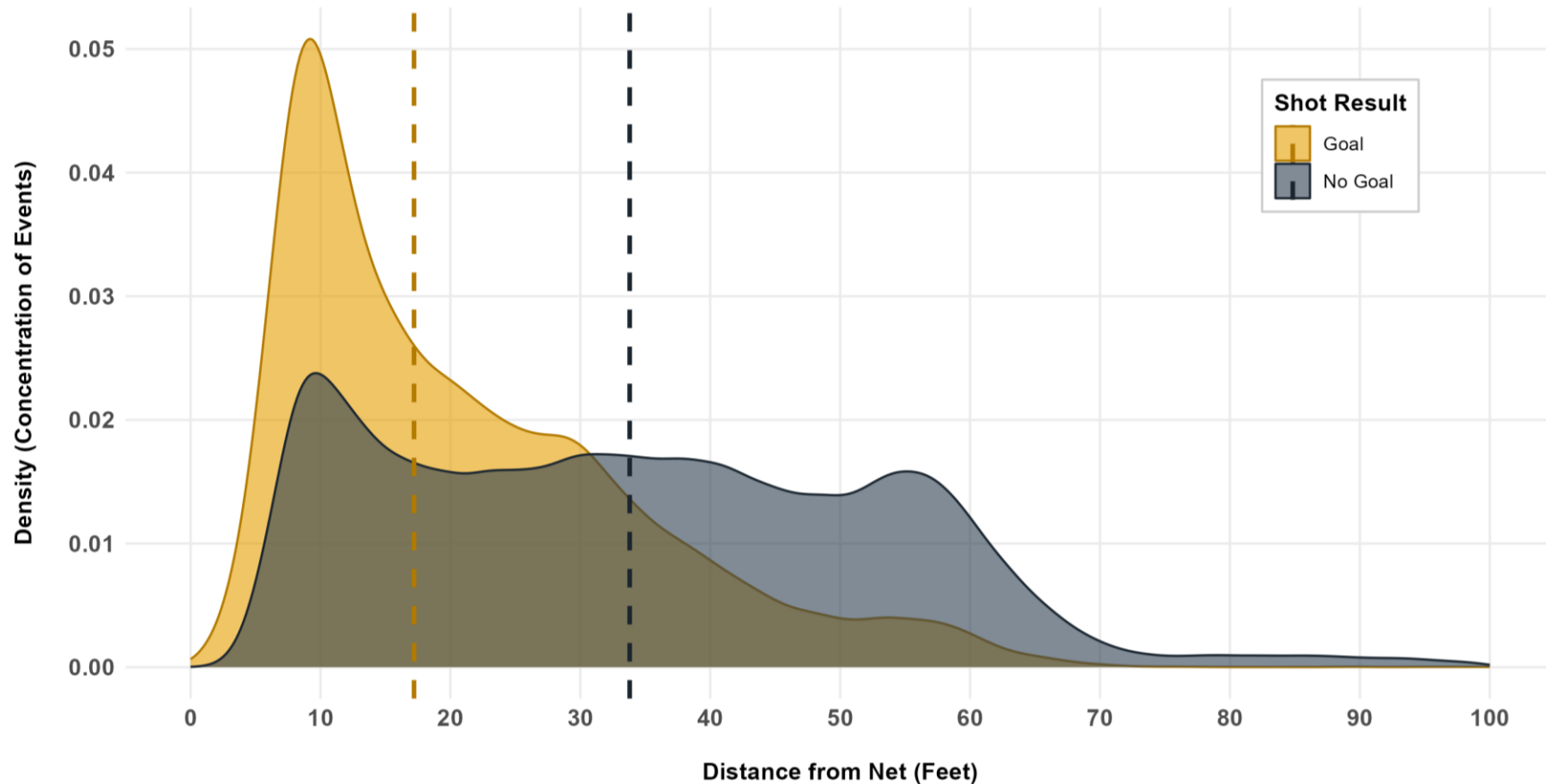
- All shots
- Peak at 10

Exploratory Data Analysis



Shot proximity dictates success: Goals occur significantly closer to the net

The median goal is scored from 17.2 feet, while the median non-goal is taken from 33.8 feet.



Data: hockeyR

- Goals peak at 10
- Non-goals are way more flat

Problem + Solution



Problem: Class Imbalance

- 91% of shots are missed/blocked
- Only 9% are goals

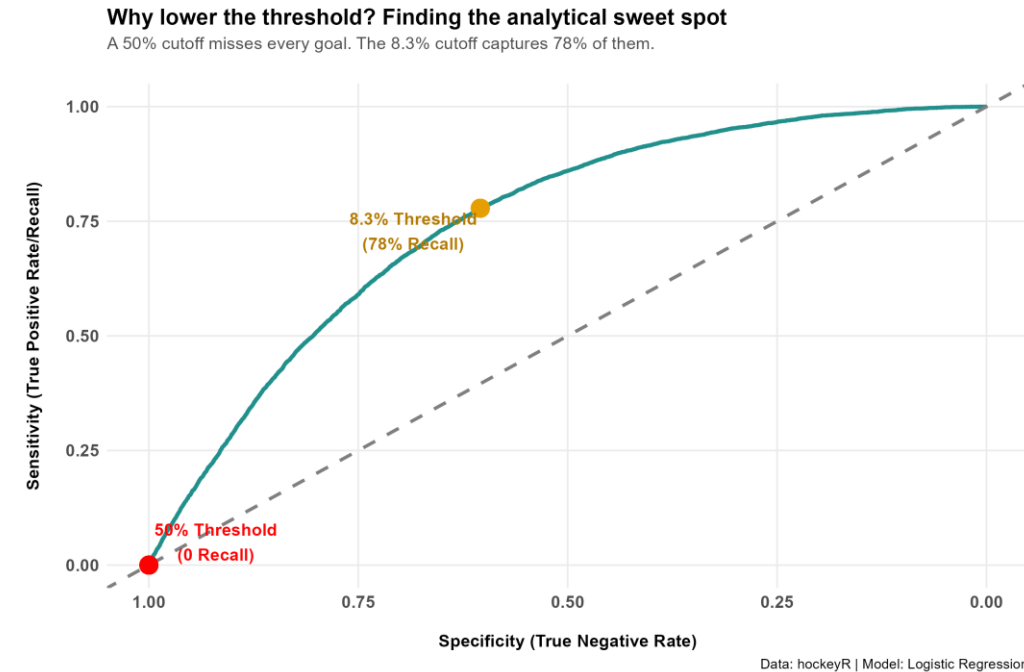
Solutions:

- Threshold Tuning
- Prior Probability Adjustment
- Balanced Downsampling

Model 1 – Logistic Regression



- Initial baseline regression
- Lowered threshold from 50% to 8.3% to prevent “no” guesses every time



Logistic Regression*

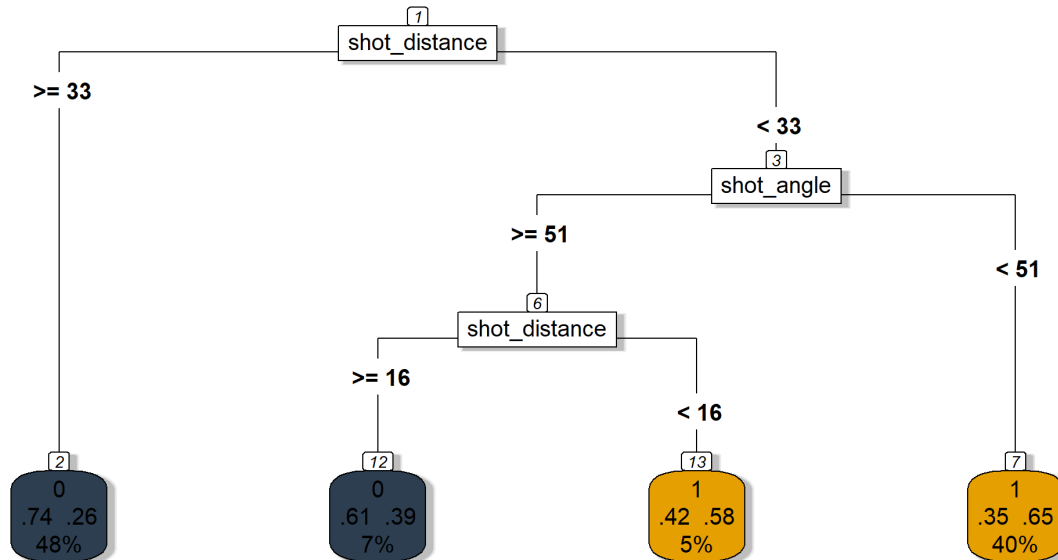
Accuracy	Precision	Recall	F-measure
61.61%	12.55%	77.84%	21.61%

*Performance on unseen 2024 data

Model 2 – Decision Tree



Final Pruned Decision Tree



- “Priors” were tuned to heavily penalize the model for missing goals
- Cross Validation was used to prune the tree to 3 splits, preventing the noise from being memorized by the model

Decision Tree*

Accuracy	Precision	Recall	F-measure
59.95%	11.96%	76.88%	20.70%

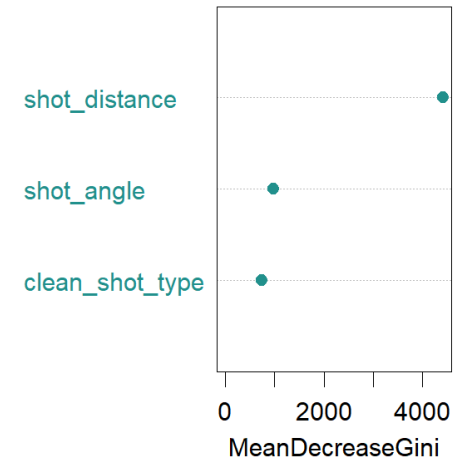
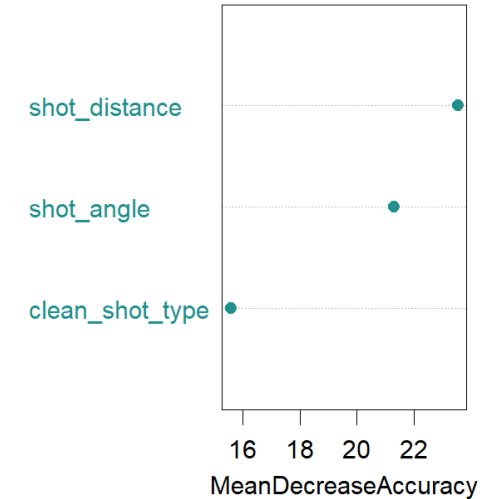
*Performance on unseen 2024 data

Model 3 – Random Forest



- 100 aggregated decision trees
- Balanced downsample dataset (30,000 goals + 30,000 saves/misses)

Random Forest: Which features drive NHL goals?



Random Forest*

Accuracy	Precision	Recall	F-measure
64.19%	12.93%	74.41%	22.03%

*Performance on unseen 2024 data

Findings and Recommendations

Systematic Adjustments for NHL Offenses

- Penetrate the inner slot
- Shots from the blue line rarely generate xG
- Net front presence → deflections + tip-ins = high xG

Missing Contextual Variables

- Wide open vs contested shots were treated equally
- No data regarding pre-shot puck movement
- No data about goalie positioning, traffic/screens, or shooter speed

The Next Evolution of xG Modeling

- Future Iterations:
 - *Use NHL EDGE player-tracking data to map defensive pressure and goaltender depth*
 - *Introduce game-state variables to account for differences in play for teams that are leading vs trailing*

Questions?



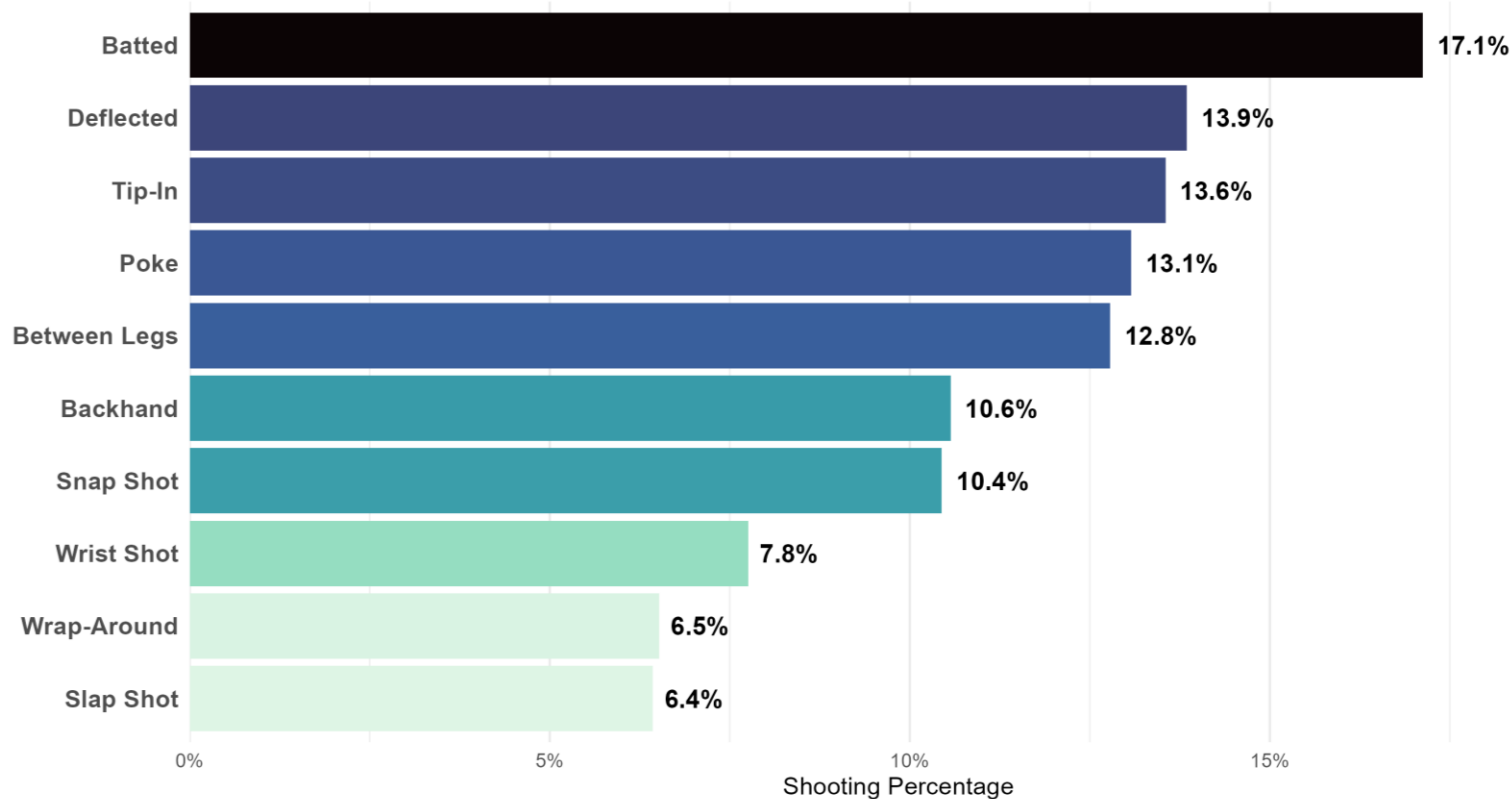
Appendix



Shot Type Conversion Efficiency

Batted pucks and redirections are the most dangerous in-game shots

Average shooting percentage by unblocked shot type (2018-24, excluding 2020-21)



Data: hockeyR

- Net front shots
- Sample Sizes

Shot Type	Total Shots	Goals
Wrist Shot	254506	19736
Snap Shot	65237	6814
Slap Shot	59059	3799
Backhand	35783	3781
Tip-In	29673	4021
Deflected	8758	1213
Wrap-Around	4111	268
Poke	872	114
Batted	660	113
Between Legs	133	17